TEAM

ASU ROAR

EUROPEAN ROVER CHALLENGE 2026

SOLO MISSION - CASE STUDY

ASU RACING TEAM

ASU ROAR

ERC EUROPEAN ROVER CHALLENGE

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Overview of the European Rover Challenge (ERC)

The **European Rover Challenge (ERC)** is an international space and robotics competition organised by the *European Space Foundation* and co-organised by *AGH University of Krakow*. Held annually in Poland, the ERC brings together student engineering teams from universities worldwide to design, build, and operate Mars-analogue rovers on a geologically realistic simulated Martian terrain known as the *Marsyard*.

The Marsyard is constructed in close collaboration with planetary geologists and faithfully reproduces geological features found on Mars, including impact craters, volcanic landforms, dune fields, and ancient water-carved valleys. Its layout changes each year to represent a different region of the Red Planet, ensuring that teams cannot rely on prior terrain knowledge.

The ERC serves as a platform for the next generation of space engineers to demonstrate technical excellence and problem-solving ability under the evaluation of experts from leading institutions — including NASA, the European Space Agency (ESA), the German Aerospace Center (DLR), and major aerospace companies such as Airbus and Thales Alenia Space.

The ERC 2026 finals are scheduled for **4–6 September 2026**.

Competition Definition

Competition Tasks

The ERC comprises five scored tasks designed to mirror real planetary exploration mission activities:

Task	Description
Presentation	Teams present their rover design, technical solutions, and lessons learned to a jury of industry experts, followed by a Q&A session.
Probing	Locate and retrieve three probes distributed across the Marsyard using the rover's manipulator; scored on completion time.
Maintenance	Operate a dedicated panel by configuring switches, measuring electrical parameters, powering an electromagnetic lock, and inserting an RJ-45 plug. Autonomous operation using ArUco markers yields a higher score.
Navigation	*Traverse*: reach five checkpoints, optionally autonomously (no video feedback) for bonus points. *Droning*: operate a drone to follow flight instructions and capture aerial photographs.
Science	Conduct geological exploration, collect surface and sub-surface samples, and deliver a scientific analysis report.

Key Scoring Drivers

Across all tasks, **autonomy** is a primary differentiator. Tasks completed without direct teleoperation or video feedback are awarded substantially higher scores. This places a heavy engineering emphasis on robust perception, , path planning, and control — the exact subsystems addressed in this assessment.

Operational Environment & Environmental Warnings

The ERC competition takes place outdoors in Poland in early September on the Marsyard — a geologically detailed terrain exposed to real atmospheric conditions. Unlike a controlled laboratory, the operating environment introduces a wide range of physical and meteorological factors that can degrade or completely disrupt autonomous system behaviour. The warnings below are treated as **hard constraints** that every subsystem must account for during design, simulation, and validation.

Figure 1: View of the ERC Mars Yard in Poland.

Note: The following is only an assist from us.

Weather Conditions

The ERC competition venue in Poland experiences variable autumn weather, including sudden rain, high humidity, gusty winds, and overcast skies, all within the same competition day.

[WARNING] Rain and High Humidity

[WARNING] Wind and Dust

Lighting and Visibility Conditions

Outdoor lighting conditions at the competition site change continuously and are outside the team's control.

[CRITICAL] Direct Sun Glare and Specular Reflections

[WARNING] Rapidly Changing Shadows

[WARNING] Overcast and Low-Contrast Scenes

Terrain Variability

The Marsyard is deliberately engineered to be challenging. Teams should not assume flat or consistent ground.

[CRITICAL] Steep Slopes and Loose Gravel

[WARNING] Irregular Rock Fields and Step Obstacles

[WARNING] Soft or Sandy Soil

Temperature Extremes

Early September in Poland can produce wide diurnal temperature swings, typically between 8 °C and 28 °C across a competition day.

[WARNING] Low Temperature and Battery Performance

[WARNING] IMU Thermal Drift

The environmental factors above collectively define the **operational envelope** of the ASU ROAR system and serve as the foundation for the corner-case analysis, simulation scenario design, and stress testing methodology described in Part 3.

ASU ROAR System Architecture

Architecture Overview

The ASU ROAR rover is a ground unmanned vehicle (UGV) built on a **ROS 2 / micro-ROS** software stack, targeting the autonomous navigation, manipulation, and scientific tasks of the ERC. The architecture follows a hierarchical decomposition: a high-level mission manager coordinates four core subsystems — Perception, , Path Planning, and Control — while a real-time embedded layer handles low-level actuation via micro-ROS.

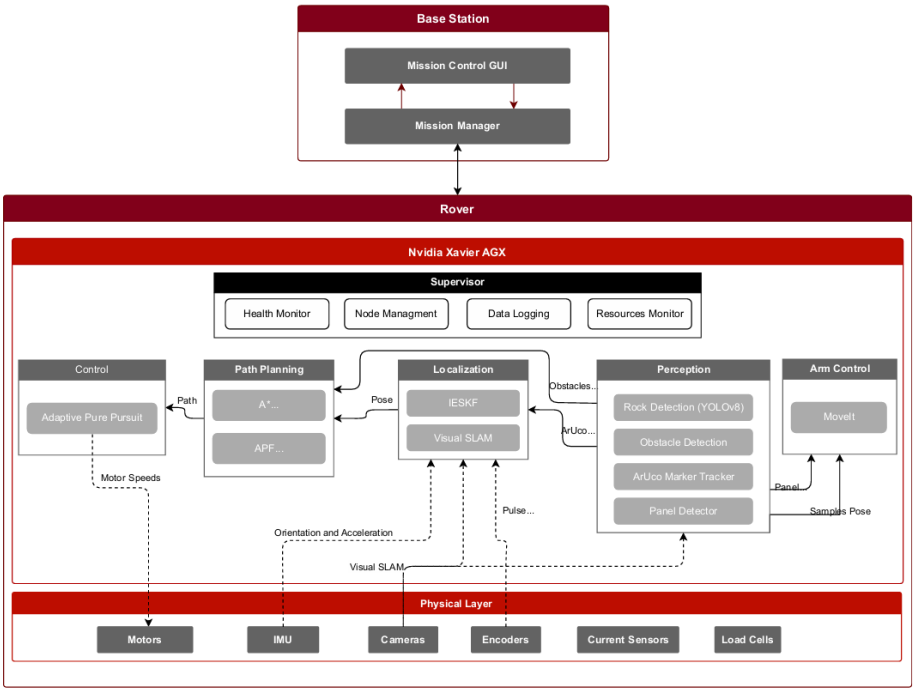

Figure 2: ASU ROAR full-stack system architecture diagram.

Sensing Layer

The sensing layer provides raw data streams consumed by both the Perception and subsystems:

- **RGB-D Camera** — visual feed used for ArUco marker detection, visual odometry, and operator monitoring.
- **IMU (Inertial Measurement Unit)** — provides linear acceleration and angular velocity at high frequency; feeds directly into the pipeline.
- **Wheel Encoders** — quadrature incremental encoders mounted on each drive motor shaft; measure angular velocity and cumulative shaft rotation, providing wheel odometry estimates of linear displacement and heading change.
- **Robotic-arm Current Sensors** — used for measuring Force/Torque during the Maintenance and Probing tasks for contact-aware manipulation.

All sensor publish to ROS 2 topics, enabling modular integration and facilitating bag-based replay for offline testing.

Communication Layer

- **micro-ROS bridge** — a dedicated micro-ROS agent running on the main computer (i.e. NVIDIA Jetson Xavier) serialises DDS messages to/from the embedded microcontroller (i.e. STM32) over UART, enabling real-time actuator commands and low-level sensor reads with deterministic latency.
- **Wi-Fi telemetry link** — provides the operator control station (OCS) with a telemetry stream and an optional video feed; used in teleoperation mode only.
- **ROS 2 DDS middleware** — all intra-process and inter-process communication on the main compute unit uses ROS 2 with a DDS transport (default: Fast-DDS). Quality of Service (QoS) profiles are tuned per topic to balance latency and reliability.

Perception Subsystem

The Perception subsystem is responsible for interpreting raw sensor data into semantic information:

- **ArUco Marker Detection** — OpenCV-based corner detection and pose estimation pipeline; outputs 6-DoF marker poses in the camera frame, used by both the Localization and Manipulation modules.
- **Obstacle Detection** — LiDAR point clouds are processed (filtered, downsampled, and clustered) to produce an occupancy grid that feeds the local planner.
- **Visual Odometry** — camera-based relative pose estimation supplements IMU data in GPS-denied regions.

Localization Subsystem

Localization fuses complementary sensor modalities to produce a reliable estimate of the rover's pose in the world frame:

- **Fuzz-IESKF** — adaptive localization algorithm that fuses wheel, IMU, and visual data to reduce navigation drift. It uses a Fuzzy Inference System to dynamically adjust sensor trust based on environmental reliability, such as lighting or wheel slippage.
- **ArUco Marker Localization** — when ArUco markers at known coordinates are detected, their pose measurements are injected as absolute position corrections, significantly reducing accumulated drift.
- **SLAM** — a Visual SLAM module (provided from the Zed2i Camera) can be activated in fully GPS-denied environments to build and localize against an online map.

Path Planning Subsystem

Path planning operates at two levels:

- **Global Planner (A*)** — given a goal pose and the global occupancy map, the A* algorithm computes a collision-free reference trajectory from the rover's current position.
- **Local Planner (APF)** — reacts in real time to dynamic or previously undetected obstacles within the sensor range, adjusting the velocity commands while tracking the global reference path.

Control Subsystem

The Control subsystem translates planned trajectories into physical actuator commands:

- **Adaptive Pure Pursuit** — computes steering commands by dynamically adjusting the lookahead distance as a function of current speed and path curvature, enabling smooth tracking on both straight and curved terrain.
- **Velocity profiling** — a speed controller enforces acceleration limits and terrain-dependent maximum velocities controlled and set by the low level sub-team.
- **Arm controller** — a separate control pipeline for the robotic arm uses PID controllers coordinated by a task-space motion planner for manipulation tasks.
- **micro-ROS actuation** — final velocity and joint commands are serialised over the micro-ROS bridge and executed by the embedded motor drivers with closed-loop PWM control.

Mission Manager

A top-level state machine (implemented as a ROS 2 node) orchestrates task sequencing, subsystem activation, mode switching (autonomous / teleoperation), and fault recovery behaviours. It interfaces with the operator control station via a ROS 2 action server, accepting high-level goal commands and streaming telemetry status.

Summary

The ASU ROAR system represents a tightly integrated autonomous stack designed around the specific task demands of the European Rover Challenge. The layered architecture — from embedded real-time control via micro-ROS up through ROS 2-based perception, , planning, and mission management — provides a principled basis for the verification and validation analysis carried out in the remainder of Part 3.

The following questions (Q1–Q7) apply this architecture as the concrete system under test, replacing the generic UGV assumption specified in the assignment with the actual design context of team ASU ROAR.